

Aptitude Preparation Notes

Section 1: Quantitative Aptitude - Percentages, Profit and Loss

How do you calculate a percentage?

Percentage equals (part divided by whole) multiplied by 100. To find x percent of a number N, calculate N multiplied by x divided by 100. For example, 20 percent of 250 is 50.

What is the formula for percentage increase and decrease?

Percentage change equals (new value minus old value) divided by old value, multiplied by 100. If the result is positive it is an increase, and if negative it is a decrease. If a price rises from 80 to 100, the increase is 25 percent.

What is the formula for profit and loss?

Profit equals selling price minus cost price. Loss equals cost price minus selling price. Profit percentage equals (profit divided by cost price) multiplied by 100. Loss percentage equals (loss divided by cost price) multiplied by 100. Both percentages are always calculated on the cost price unless stated otherwise.

Example: An article is bought for 500 and sold for 600. Find the profit percent.

Profit is 600 minus 500, which is 100. Profit percent is 100 divided by 500 multiplied by 100, which equals 20 percent.

What is simple interest and compound interest?

Simple interest equals principal times rate times time divided by 100. Compound interest amount equals principal multiplied by $(1 + \text{rate} / 100)^{\text{time}}$. Compound interest is earned on both the principal and the previously earned interest, so it is larger than simple interest for periods over one year.

Section 2: Quantitative Aptitude - Time, Speed, Work and Ratio

What is the relation between speed, distance and time?

Speed equals distance divided by time. Distance equals speed multiplied by time. Time equals distance divided by speed. To convert km per hour to metres per second, multiply by 5 and divide by 18.

How do you find average speed for equal distances at two speeds?

When the same distance is covered at speeds a and b , average speed is not the simple average. It is the harmonic mean, which equals $2ab$ divided by $(a + b)$. For 40 km/h and 60 km/h the average speed is 48 km/h.

How do you solve time and work problems?

If a person completes a job in n days, the work done in one day is 1 divided by n . If A and B work together, their combined one-day work is the sum of their individual one-day work. If A takes 10 days and B takes 15 days, together they take 6 days, because $1/10$ plus $1/15$ equals $1/6$.

What is ratio and proportion?

A ratio compares two quantities of the same kind, written $a:b$. A proportion states that two ratios are equal, $a:b = c:d$, which means a times d equals b times c . To divide an amount in the ratio $a:b$, the first share is a divided by $(a + b)$ of the total.

What is the formula for average?

Average equals the sum of all observations divided by the number of observations. If the average of 5 numbers is 20, their sum is 100.

Section 3: Logical Reasoning

What are the common types of logical reasoning questions in placement tests?

Common types include number series, letter series, coding and decoding, blood relations, direction sense, seating arrangement, syllogisms, and puzzles.

How do you solve number series questions?

Look for a pattern in the differences between terms, such as a constant difference, an increasing difference, multiplication by a fixed number, squares, cubes or alternating patterns. For the series 2, 4, 8, 16, the next number is 32 because each term is double the previous.

How do you solve blood relation questions?

Draw a family tree using symbols for male and female and connect people with the relations given. Work step by step from the information in the problem instead of guessing, and note that terms such as maternal and paternal show which side of the family a person belongs to.

How do you solve direction sense questions?

Draw the path on paper. Mark north at the top, and track each turn. A right turn from facing north leads to facing east. Use the Pythagoras theorem to find the straight-line distance between the start and end points when the path forms a right angle.

What is a syllogism?

A syllogism is a type of reasoning question with two or more statements and conclusions. You must decide which conclusions follow from the statements, regardless of common knowledge. Venn diagrams are the easiest way to check each conclusion.

Section 4: Verbal Ability and Data Interpretation

What topics are covered in verbal ability?

Verbal ability covers reading comprehension, synonyms and antonyms, sentence correction, fill in the blanks, para jumbles and error spotting. Regular reading and vocabulary practice improve performance.

How do you handle reading comprehension questions?

Read the questions first, then read the passage while looking for answers. Focus on the main idea and the author's tone, and do not rely on outside knowledge. Eliminate the options that are clearly wrong.

What is data interpretation?

Data interpretation questions present data in tables, bar charts, line graphs or pie charts and ask you to calculate or compare values. Common skills are percentages, averages, ratios and growth rate.

What tips help in aptitude tests?

Learn the key formulas, practise mental maths, solve timed mock tests, attempt easier questions first, avoid spending too long on a single question, and review your mistakes after every test.